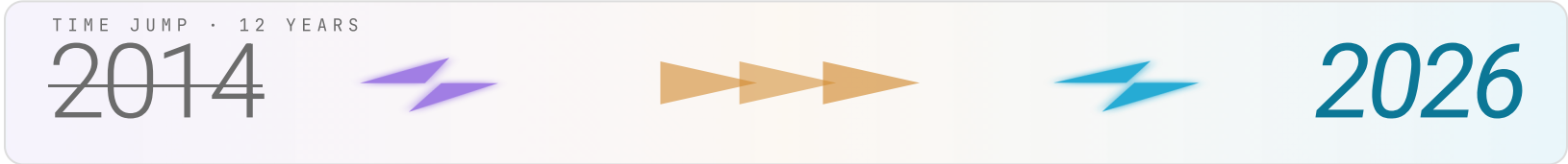


Where we're going, we don't need a *monolith*.



jQuery was still cool when this codebase was born. This cycle the monolith stepped down — replaced by a Module-Federated mesh and a portable API layer.

Jean-Claude Kirwan · Frontend, Tradify · TRA-22640 ▶ 88 MPH

TIME CIRCUITS · ON

DESTINATION TIME	CYCLE 02 · EXEMPLAR <i>post-Spark · pillars in production</i>
PRESENT TIME	06 MAY 2026 <i>Spark · Cycle 01 wrap</i>
LAST TIME DEPARTED	23 MAR 2026 <i>cycle kickoff</i>

POWER DRAW

1.21 GW · *a bolt of lightning*

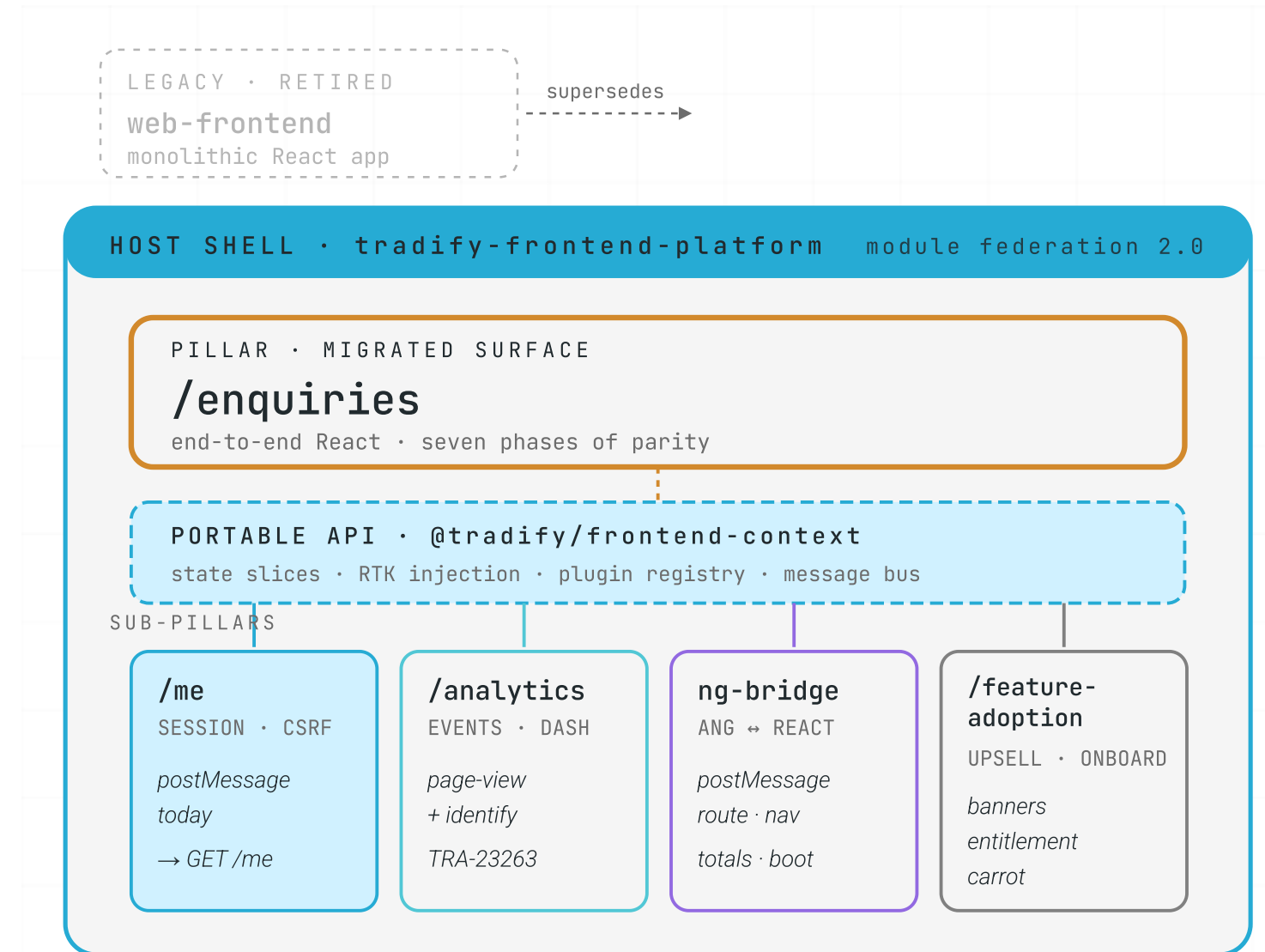
INSIDE

Foundation · what got built · *Challenges* · obstacles faced · *Enhancements* · sharpened the saw

Where we're going, we don't need *jQuery*.

The legacy `web-frontend` — born when jQuery was still cool and `$scope.$apply()` was a love language — stepped politely off the timeline. In its place, `tradify-frontend-platform`: a Module-Federation host that *contains* every MFE and lets them speak through one portable API layer, `@tradify/frontend-context`. Great Scott!

- **/enquiries** — first surface migrated end-to-end. Seven phases of parity with the Angular original; the proof the platform is real. Stands on three sub-pillars:
 - **/me** — session, CSRF, identity. Bootstrap today; `GET /v2/api/v1/me` tomorrow — same contract, zero consumer churn.
 - **/analytics** — page-view telemetry, dev dashboard, parity with Angular events (TRA-23263).
 - **AngularJS message bridge** — `postMessage` choreography between the legacy shell and React: bootstrap, route sync, navigation interception, sidebar totals.
 - **/feature-adoption** — upsell banners & onboarding panels; the carrot when an entitlement gate denies access.



An old *dog*, four new tricks.

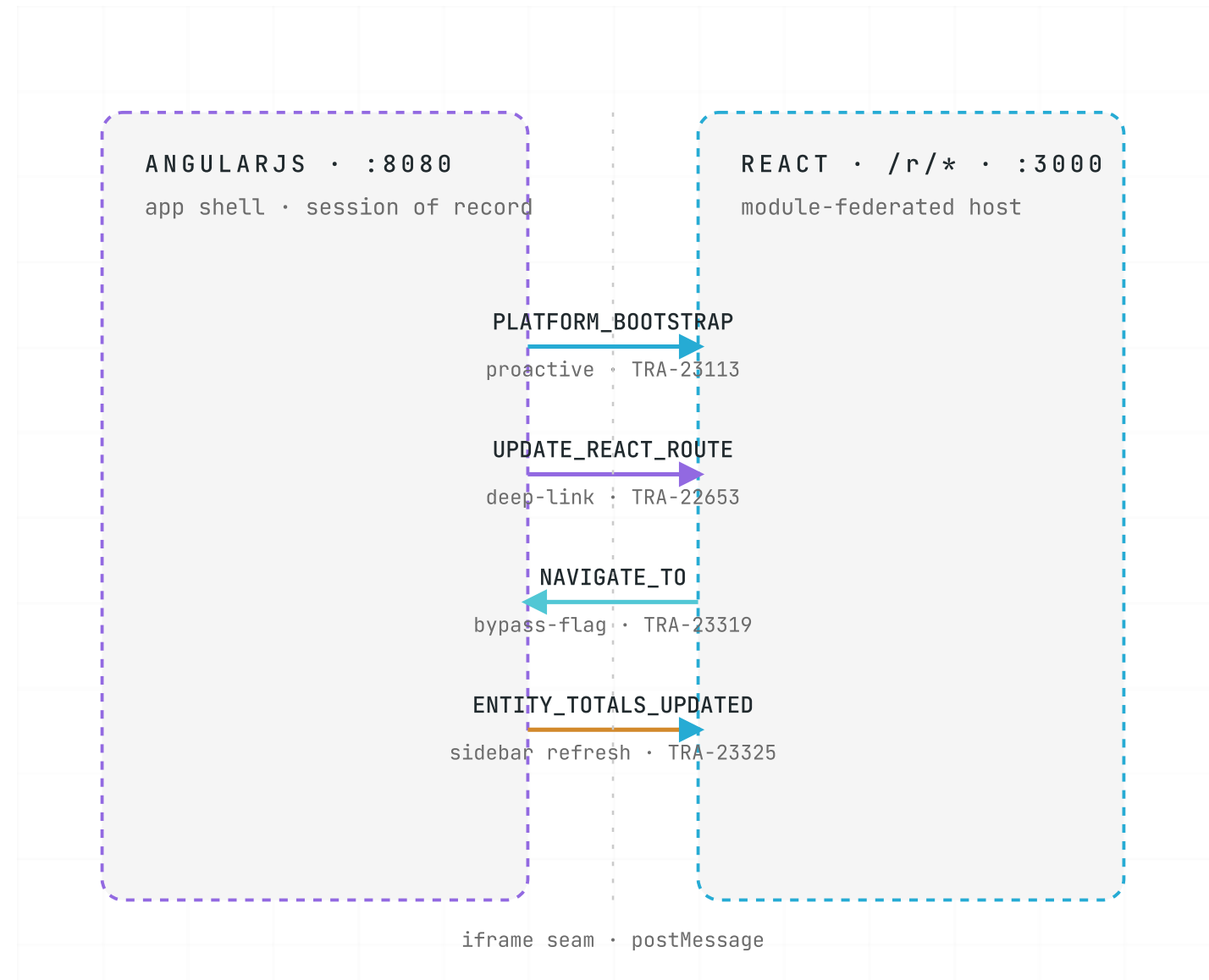
The 2014 shell still hosts production. Picture a pile of *444-line* controllers, the smell of `$scope` in the morning. This cycle that shell learned to broadcast bootstrap proactively, intercept cross-frame nav, and stop fighting the React router on its own page — without ripping a single `ng-controller` out.

Angular → PLATFORM_BOOTSTRAP — *session, csrfToken, identity* TRA-23113

Angular → UPDATE_REACT_ROUTE — *deep-link sync* TRA-22653

React → NAVIGATE_TO — *bypass-flag for inner routes* TRA-23319

Angular ↔ PLATFORM_ENTITY_TOTALS_UPDATED — *sidebar count refresh* TRA-23325



Two *leaks*. Both sealed.

α PERFORMANCE · 1.21 GW · ENQUIRIES COLD-LOAD

The *six-second* white screen.

~~6.0s~~ → 1.9s

A waterfall straight out of 2014: source aliases, eager-shared leaks, a CDN-fetched manifest. Collapsed into one inlined manifest, a preloaded remote entry, and a host-shell hoist of `/me`. Roads? Where we're going we don't need round trips.

TRA-23095	Removed source aliases from plugin <i>boundaries</i>
TRA-23113	Proactive <i>PLATFORM_BOOTSTRAP</i> push
TRA-23293	Host-shell <i>/me</i> hoist + manifest
TRA-23325	Killed polling + tab-change <i>refetch</i>
perf #59	Eager-shared <i>follow caller</i>

β ENTITLEMENT GATES · PLAN-TIER PADLOCKS

Doc Brown's *padlock*, ported.

AngularJS shipped `<upsell-padlock>` — a plan-tier gate riveted onto the 2014 dashboard. Re-implementing it in an MFE that boots *before* bootstrap arrives surfaced two paradoxes at once: a contract the platform didn't have, and a loading state nobody wanted to see.

```
<EntitlementGate
  entitlement="purchaseOrders"
  fallback={<FeatureAdoptionPanel />}
  optimistic
>
  <PurchaseOrderList />
</EntitlementGate>
```

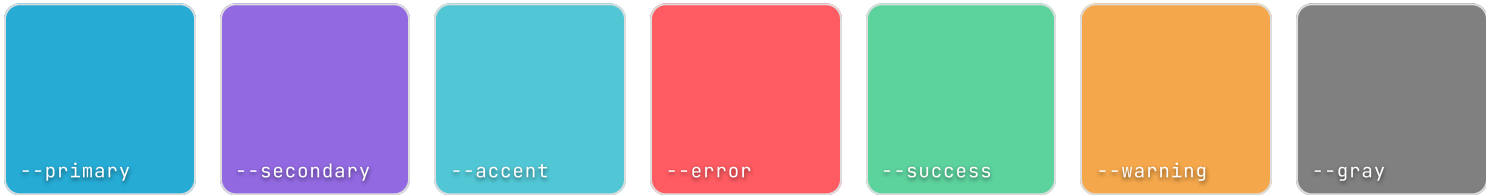
useEntitlement — reads `bootstrap.entitlements[key]` off the `/me` slice; resolves **GRANTED** · **DENIED** · **LOADING**

optimistic — render children while bootstrap is in flight; swap to fallback if denied. Trades a brief flash for happy-path speed (off for paid-only content).

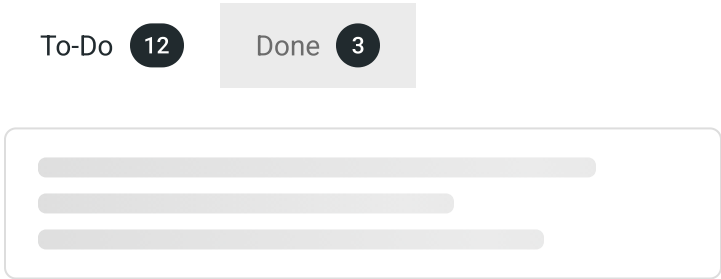
FeatureAdoptionPanel — the carrot inside the padlock: heading, body, primary CTA, dismiss/remind. The upsell when access is denied *or* the onboarding banner when the feature exists but isn't set up yet.

One *flux capacitor*, two power sources.

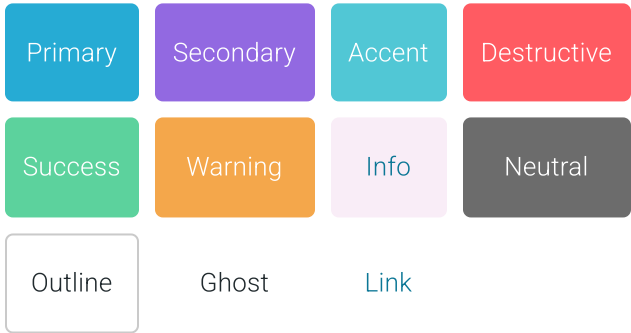
Every component reads from token contracts under `[data-brand]` — Tradify-blue ↔ Access teal flips at runtime. Hit **B** on the keyboard. Plutonium not required.



TABS · STORYBOOK PARITY



BUTTONS · STORYBOOK PARITY



PARITY · TRA-23228

Parity, audited.

ui-components aligned to Evo specs across button, badge, tab, modal, tooltip — without forking. The contract is the seam.

TRA-23228	Evo brand parity for ui-components
TRA-23083	Brand theming consolidation (review)
TRA-22952	Tooltip styling contract
TRA-23163	Standardised loading spinner position
—	Access · Evo integration handbook in Storybook

A bench of *Docs*, on call.

Stood up **tradify-agent-skills** — the user-level garage for tech-lead agents and project skills, shared across every repo. Each one a Doc Brown for its domain. Repo-level agents handle bundle, platform, and DevX work close to the code — no DeLorean required.

USER · TECH LEADS · 8
tradify-agent-skills/agents

frontend-tech-lead	React 19 platform, MF, Tailwind
backend-tech-lead	.NET 10 / 4.7.2, EF, multi-tenancy
angularjs-tech-lead	legacy shell, interop, digest cycle
ux-tech-lead	Figma · Evo · WCAG · component patterns
e2e-testing-tech-lead	cross-repo coverage strategy
devops-engineer	Docker, Azure, CI/CD pipelines
security-reviewer	tenant isolation, CSRF, OWASP
engineering-manager	cross-repo arch & impact

REPO · SPECIALISTS · 7
.claude/agents

platform-engineer	Rspack · MF · plugin manifest
frontend-context-architect	portable API · RTK injection
frontend-architect-ui	ui-components · Storybook · a11y
bundle-performance-engineer	chunk graph · waterfall · LCP
devx-engineer	generators · Turbo · scripts/
docs-engineer	READMEs · ADRs · runbooks
code-reviewer	DRY · dead code · package-agnostic

SKILLS · PROJECT-SPECIFIC · 7
.claude/skills

evo-shadcn	token theming · ui-components rules
module-federation	host/remote · shared singletons
react-19	R19.2 pinned · no SSR · Tailwind-only
tailwind-v4	token contracts · no inline styles
html5-react	JSX semantics · WCAG · ARIA
storybook-10	CSF Factories · webpack5 builder
cross-platform-dx	macOS + Windows scripts & paths

Hoverboard-grade developer loops.

GENERATORS · SCAFFOLD FROM ONE COMMAND

- `generate:mfe` — Module Federation plugin from template
- `generate:evo-mfe` — Evo-themed MFE w/ tokens wired
- `generate:context` — frontend-context state slice
- `generate:api-call` — RTK Query endpoint w/ types
- `generate:icons` — Lucide → ui-icons SSR-safe

TARGETS & MODES · SAME SHELL, MANY BACKENDS

- `dev` — local stack w/ Docker proxy & Angular shell
- `dev:qa` — point at QA APIs from local React
- `dev:mock` — frontend-context mock bootstrap
- `switch-api.mjs` — flip targets without restart
- `ensure-env-local.mjs` — fail-soft .env bootstrap

CONTRACTS & PARITY · CATCH DRIFT IN CI

- `check:brand-parity` — tokens align across brands
- `validate-shared-config` — Rspack singletons match
- `validate-manifest-consistency` — plugin manifest schema
- `validate-env-completeness` — every MFE has env keys
- `sync:singletons` — propagate React/RTK pins

DEV-ONLY VIEWS & AGENT MODE

- `/analytics` dashboard — non-prod visualisation of events
- `/feature-adoption` playground — banner variants live
- generators support `--agent` for non-interactive use
- `graph` script — render dependency graph as SVG
- Design-Sync DevX branch — staged for merge

Demo time.

The *almanac* doesn't lie.

UNIT TEST COVERAGE
LINES COVERED · COMBINED

95.1%

4,313 / 4,537 lines

PERF · OUT OF IFRAME
PROJECTED · REACT-OWNED ROUTE

~0.8s

no shell relay · no postMessage hop

A11Y
COMPONENTS A11Y-ATTRIBUTED

36 / 49

aria · roles · focus mgmt

QA
/ENQUIRIES TEST FILES

24

6,548 lines of spec

CI
TURBO AFFECTED-GRAPH

~5x

build & test only what changed

ARCH
NEW MFES · BRAND SURFACES

4 · 2

/me · /analytics · /FAP · /enquiries

SAME FEATURE · TWO ARCHITECTURES · ONE ROOF VS INJECTED ENDPOINTS

```
// 2014 · one roof · EnquiriesCtrl.js
angular.module('tradify')
  .controller('EnquiriesCtrl', function(
    $scope, $http, $state,
    SessionSvc, ToastSvc, ...
  ) {
    $scope.enquiries = [];
    $scope.load = () => $http.get(...);
    $scope.save = (e) => $http.post(...);
    $scope.del = (id) => $http.del(...);
    $scope.format = (row) => { ... };
    $scope.valid = (e) => { ... };
    $scope.edit = ($i) => { ... };
    // + 380 lines: auth, paging, DOM, errors
  });
// state + api + format + nav — entangled.
// every feature reinvents auth, retry,
// cache, invalidation. god-controller.
```

```
// 2026 · feature owns its slice · enquiriesApi.ts
export const enquiriesApi =
  baseApi.injectEndpoints({
    endpoints: (build) => ({
      list: build.query<Enquiry[], Args>({
        query: (a) => ({ url: '/enquiries' }),
        providesTags: ['Enquiries'],
      }),
      save: build.mutation<Enquiry, In>({...}),
      remove: build.mutation<void, string>({...}),
    }),
  });
export const { useListQuery, useSaveMutation }
  = enquiriesApi;
// baseApi gives auth, retry, cache,
// invalidation — for free.
// feature only declares what's NEW.
// composable, testable, code-split per MFE.
```

1 controller, 444 lines → N features, each owning ~80 lines · auth/retry/cache solved once in baseApi
· add a feature = injectEndpoints, not edit a god-controller